ADITYA J. VALAKI

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PROFILE

Final-year B.Tech Information Technology student (CGPA 8.17/10, Graduation Completing on June 2027) focused on applied machine learning, statistics and data analytics. I led a controlled benchmark of 11 AI reading engines for a medical-AI research project, where Gemini 3.5 Flash reached 86.1% F1 and the ranking was backed by Wilcoxon signed-rank significance testing rather than raw averages. I have built deep learning and NLP classifiers, a Python pipeline over high-frequency Formula 1 telemetry for race-strategy modelling, and business dashboards for non-technical stakeholders. Worked as Sales and Marketing Head in a software company and two business-facing internships taught me to scope a vague question, test the answer properly and explain it plainly. Research output includes an IEEE-format conference paper and a journal submission.

TECHNICAL SKILLS

Programming: Python, SQL, C++, C#, JAVA, JavaScript, HTML/CSS

Machine Learning & Deep Learning: scikit-learn (Random Forest, ensemble methods), TensorFlow/Keras, CNNs, transfer learning (EfficientNetB0), NLP (text preprocessing, TF-IDF vectorisation), supervised classification and regression, model evaluation (precision, recall, F1, confusion matrix)

Generative AI & LLMs: multimodal LLM / vision-language model benchmarking (Gemini, GPT-4o, Claude, on-device VLMs), LLM API integration, prompt engineering, OCR (Tesseract), LangChain, Deep Agents.

Data & Statistics: pandas, NumPy, exploratory data analysis, feature engineering, data cleaning, time-series and telemetry data, hypothesis testing (Wilcoxon signed-rank), correlation analysis, fuzzy string matching (RapidFuzz)

Visualisation & BI: matplotlib, seaborn, Power BI, IBM Cognos, Excel, Streamlit dashboards

Engineering & Tools: FastAPI, React Native (Expo), React, Next.js, Supabase, Firebase, FastF1, Joblib, Git/GitHub, Jupyter; cloud computing fundamentals (NPTEL)

EXPERIENCE

Summer Research Intern — Dhirubhai Ambani University (DA-IICT), Gandhinagar *May 2026 – Jul 2026*

Applied AI research group, under Prof. Dr. Anupam Rana

- **Benchmark design:** designed and ran a controlled benchmark of 11 AI reading engines (7 cloud multimodal LLMs, 3 on-device vision-language models, 1 Tesseract OCR baseline) for handwritten Indian prescription extraction, on 50 real, pharmacist-verified prescriptions covering 192 medicines.
- **Evaluation pipeline:** built the Python scoring pipeline with RapidFuzz fuzzy matching and per-item precision, recall and F1; ran Wilcoxon signed-rank tests on per-prescription F1 so the ranking held up statistically.
- **Result:** established Gemini 3.5 Flash as the top engine at 86.1% F1 (90.1% recall) and showed on-device models were not viable (<24% F1), which ruled out an offline-only design and settled the product architecture.
- **Failure analysis:** categorised every misread and isolated 11 medicines no model could read, which made human verification and confidence gating mandatory steps in the system design.
- **Output:** authored a 6-page double-blind IEEE-format conference paper and a formal internship report documenting methodology, benchmark and results.

Business Development Executive Intern — Zeeniith Technology *Dec 2025 – Aug 2026*

zeeniith.in — *software and web development services*

- Owned the commercial side of the business end to end: lead sourcing, client outreach, pricing, proposals and delivery follow-through.
- Tracked pipeline and outreach data to decide where sales effort went, and reported conversion and delivery status back to the team.
- Also, ScholarPlace (scholarplace.in), an AI-integrated B2B edtech platform for engineering colleges; built its prospect list and segmented targets for outreach.

Business Analyst Intern — Cognifyz Technologies, Nagpur, Maharashtra *May 2025 – Jul 2025*

- Performed exploratory data analysis on a financial investment dataset in Python (pandas, matplotlib, seaborn).
- Identified patterns in investment preference, risk appetite and savings behaviour across investor segments.
- Ran correlation analysis (e.g. age vs. investment duration) to explain the behavioural drivers behind the headline numbers.
- Built visual dashboards communicating ROI expectations and decision drivers, and presented data-driven recommendations on investment strategy to non-technical stakeholders.

Business Administrator Intern — Euro Foods Industries, Mahuva, Bhavnagar *Dec 2024 – Jan 2025*

- Coordinated client communication across the sales and marketing departments, improving the response workflow.
- Supported execution of promotional campaigns and customer outreach activities.
- Assisted negotiation and procurement decisions within supply operations, and worked with cross-functional teams on day-to-day operations.

PROJECTS

1. Formula 1 Race Strategy & Telemetry Analytics | Python, FastF1, pandas, NumPy, scikit-learn, matplotlib

- Built a Python ETL pipeline on the FastF1 API (cached session loading) to ingest lap timing, tyre stint and multi-channel car telemetry (speed, throttle, brake, gear, RPM) across a full season.

- Characterised the dataset in big-data terms — volume, velocity (sampling rate) and variety of sources — to choose the storage and processing approach.
- Cleaned lap data by filtering in/out laps, safety-car laps and statistical outliers, then fitted per-compound tyre degradation curves (lap time vs. tyre age regression).
- Quantified undercut success rate by circuit by comparing rival gap deltas before and after pit-stop cycles.
- Engineered stint features (tyre age, compound, lap-time trend, track) and trained a supervised regression model to predict laps remaining until the next pit stop — a remaining-useful-life style estimate.
- Presented the work as my Big Data Analytics academic poster, turning the modelling into one visual story for a non-specialist audience.

2. Medication AI – Handwritten Prescription Reader & Reminder App | Python, FastAPI, React Native (Expo), RapidFuzz, LLM APIs

- Engineering build on top of the 11-engine benchmark (see Experience): the winning multimodal LLM powers extraction behind a FastAPI backend.
- Built a fuzzy-matching correction layer that snaps extracted names to known medicines, plus confidence-based verification routing so low-confidence items are never saved silently.
- Co-developed the React Native (Expo) app: prescription capture, a confirm-each-medicine verification UI, dosage-aware reminder scheduling and one-tap SOS calling.

3. Visual Malware Detector v3 | Python, TensorFlow/Keras, scikit-learn, Streamlit, Joblib, hashlib | GitHub

- Hybrid three-layer detection framework: an Authenticode-aware Shannon-entropy pre-filter, an EfficientNetB0 CNN (transfer learning) on grayscale byte-plot images from the Malimg dataset (25 malware families), and a Random Forest on 69 PE-header features (ClaMP).
- Combined model outputs through weighted score fusion for more robust family prediction than either model alone.
- Deployed as a real-time Streamlit app: file hashing (hashlib), serialised models (Joblib), and a family prediction with confidence score per upload.

4. FakeShield – Fake Content Detection System | Python, NLP, scikit-learn | GitHub

- NLP pipeline for misinformation detection: text cleaning, tokenisation, stop-word removal and TF-IDF vectorisation feeding supervised classifiers.
- Output framed as a credibility score rather than a hard label, so borderline content is flagged for human review.

5. SmartSpend – Personal Finance Tracker | HTML, CSS, JavaScript, Python | smartspend.dev

- Student expense tracker with local-storage income/expense recording, category-wise aggregation and budget-limit logic; structured data model ready for predictive expense analytics.

RESEARCH & PUBLICATIONS

- **Journal (under submission):** Hybrid malware detection with EfficientNetB0, Random Forest and an entropy pre-filter — Journal of Universal Computer Science (J.UCS), Austria; Scopus- and Web of Science-indexed.
- **Conference paper:** Benchmarking 11 AI reading engines for handwritten prescription extraction — 6-page double-blind IEEE-format paper prepared for IEEE DELCON 2026.
- **Academic poster:** Formula 1 telemetry analytics and race-strategy modelling — Big Data Analytics, Semester 7.

EDUCATION

B.Tech, Information Technology - G H Patel College of Engineering & Technology, CVM University *Jun'23 – Present(Completing Jun'27)*
Vallabh Vidyanagar, Gujarat | CGPA 8.17 / 10, no active backlogs | Coursework: Machine Learning, Big Data Analytics, Data Science & Visualisation, DBMS, Statistics, Cloud Computing

Higher Secondary (Science – Mathematics) — Shree Gyanmanjari Vidyapith, Bhavnagar *2021 – 2023*
74.75% | Secondary: Gyanmanjari Vidyapith, Bhavnagar — 90.20% (2021)

LEADERSHIP & ACTIVITIES

- IEEE Gujarat Section — R10 ACEI Entrepreneurship Ambassador (May 2025 – Jul 2025).
- IEEE GCET Student Branch — Joint Treasurer and Branch Coordinator (Jan 2025 – Jun 2025).
- E-Cell GCET — manage entrepreneurship cell activities.
- ISTE GCET SB — Sub-Core Member (Jan 2024 – Mar 2025).

CERTIFICATIONS & TRAINING

- IBM Data Analyst Professional Certificate (Coursera); IBM Data Visualization and Dashboards with Excel and Cognos.
- NPTEL (IIT Kharagpur) — Introduction to Internet of Things; Cloud Computing (in progress).
- SVNIT x Pantech — AI-driven software development internship; SVNIT x Remark Skill — two-day hands-on Data Science workshop.
- IEEE Code-e-Fiesta 2.0 (Python) and 3.0 (AI-assisted coding with Copilot);
- MATLAB, Simulink and Simscape Battery Onramp.

INTERESTS

Watch Formula 1 and derive strategy-telemetry (I keep my own race-by-race notebooks), Travelling, Bike Riding.

LANGUAGES KNOWN

English (full professional proficiency), Hindi, Gujarati.